

2 AI, Crypto Assets, and Financial Markets

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2.1 Introduction

Globalization brings a reduction in the barriers—whether technological or legislative—to economic exchanges between countries. Moreover, globalization brings the gradual integration of economies and societies driven by novel technologies, new economic interaction, and interrelation among the national and international policies of various players such as governments, international organizations, business, labor, and civil society. Hence, globalization increased the cross-border flow of capital, goods, services, and information.

Both domestic and cross-border M&A deals are susceptible to insider trading for the reason that acquirers traditionally pay a large premium for target stocks. Thus, traders who acquire target stocks or call options before the public announcement of a deal often stand to earn large profits upon the announcement. Hence, in a cross-border M&A, from the target country regulators' standpoint, insiders linked with the foreign acquirer are mostly residents of foreign jurisdictions, which means that for these foreign insiders, while their illegal trades take part in the target country's market, the crucial evidence required for prosecuting them is in most cases not under the control of the target country authorities. Thus, an executive in the acquirer can easily tip a friend regarding the forthcoming cross-border M&A deal. To that extent, Guo, Lixiong, and Zhang, Sherry¹

find that “cross-border deals involving acquirers from countries with weak legal institutions, more acceptance of cheating on taxes, and high corruption are associated with

greater intensity of insider trading. This provides further evidence that some of the trades on the target country's financial market are driven by inside information leaked from foreign acquirers.”

Finance is vital for strong entrepreneurship and, subsequently, constraints on finance have an impact. Also, finance holds back the opening of markets along both supply and demand lines through its influence on the growth of fixed-cost investments such as infrastructure.

The financial markets have enlarged and altered significantly owing to enhancement in technology and the ever-altering regulatory and social environment. Moreover, there has been a spread of financial notions, instruments, and tools because of augmented computing power and immediate information sharing through networks. Hence, the globalization of businesses and economies continues to affect the field of finance, and so, in an AI economy, investors foresee information shifts by adapting their portfolios to defend against these shifts.

Advances in digitization and information technology have profoundly altered and transformed the financial industry; as a result, traditional financial institutions and banks are losing their sovereignty as new market entrants and emerging technologies replace or substitute their part as financial intermediaries.

2.2 Digitalization

First of all, digital technologies have spread into economic activity, altering production patterns via automation and consumption patterns via increasingly digital platforms. Moreover, the principle of digital technology is the “binary-digits” setup for storing information and encompassing processing, storage, and exchange of data. It is worth mentioning here that the ongoing digitalization is leading to rapid transformations in the whole structure of the global economy.² Two driving forces are obvious: firstly, digital platforms as a dominant business model, and second, an expanding role for intangible inputs such as data and software. Hence, digitalization has caused the transformation of the global economy into the digital

economy. Thus, further than the control of the economy by digital platforms, an augmented function for intangible inputs such as data and software is a quality of the digital economy which means that by bringing forth noteworthy scale economies with amplified fixed costs and diminished marginal costs benefitting large corporations³ are offering a competitive edge for early technology adopters. On the other hand, digitalization enhances productivity in the short run by deterring entry by new companies, leading to lower growth in the long run.⁴ To that extent, S. Abis and L. Veldkamp⁵ find that a considerable transformation in the production function has taken place, and the adoption of AI and big data technologies, along with the collection of stocks of data, diverges broadly in corporations' manufacturing progress.

It could be said that digital technology allows developing countries to augment economic growth and thus reduce the difference with developed countries by enabling them to go beyond physical constraints in infrastructure, which means boosting integration and stimulating national competitiveness globally. In line with this, Thuy Linh Cao and Judy Hsu⁶ argue that "digitalization can even effectively reduce the detrimental effects of geographic and religion gap on exports. Besides, digitalization is also seen to help countries overcome language, administration, and economic gaps to promote export activities."

Digital technology has let corporations to boost their product and service reach globally, capturing opportunities to augment operational effectiveness, including managing productivity and customer relationship management (CRM).⁷ Moreover, the integration of AI technologies into international business practices is indispensable for corporations to remain competitive, which means that corporations have to adapt to and acclimatize in the AI global landscape.

Digital transformations change microeconomic allocation via the electronic functioning of markets and macroeconomic outcomes, generating alterations in integration and globalization. Moreover, digitalization has fallout effects on modern economies and societies, such as notable savings in costs.⁸ It seems that current digitalization has not been transforming the central laws of economics but has activated alterations concerning agents'

interaction in the market. To that extent, V. Labhard and J. Lehtimäki⁹ suggest that

“better institutions and governance tend to be associated with greater growth-enhancing effects from digital technologies... while technology has historically had a growth-enhancing effect, the absence or presence of established and high-quality societal environment can help in amplifying the potential growth effects”.

While digital technologies influence sectors in different ways, they generate productivity gains at the corporate level, with more productive corporations being more digital.¹⁰ To that extent, digitalization generates digital goods and services.

It has to be taken into consideration that information facilitation and management have a central role in market running, demanding the cooperation of public and private sectors. To that extent, Salomé Viljoen¹¹ argues that

“the market machine mechanisms deployed in digital platforms both vex and challenge these practices, exposing contradictions between digital markets and the grounding theories that justify market deference. Yet this contradiction presents not only a regulatory challenge, but an opportunity to consider how informationalism beyond managerialism may be put toward a different regulatory approach and help to realize a different set of allocative goals and governing institutions”.

Hence, information procedures allow digital platforms to adjust prices to their benefit, but on the other hand, high-quality public data permit institutional designs that transfer power toward community control.

It is distinctive that digital technologies are very general, applying all over the economy and activities. Moreover, digital technologies have characteristics that make them more authoritative than other technologies,

making them more difficult to fully take advantage of. Additionally, the significance of most new technologies comes from the fact that they spread across many different users and uses. Is there an interplay of digital technologies with institutions and governance? Institutions play a significant part in forming modern economies, being crucial drivers of growth.¹² Hence, digitalization, institutions, and governance stipulate openings to economic agents regarding information access, a framework for market transactions, and by permitting enforcement. In other words, digitalization by utilizing AI systems allows the SEC to spot and prevent insider trading.

It is worth mentioning here that even though digitalization is regarded as a massive game changer that would accelerate corporations' productivity and produce value for the economy as a whole, this has not transpired yet. Robert Anderton et al.¹³ show that

“digitalisation tends to have on average a positive impact on firm level productivity growth. However, it does not seem to be either a massive gamechanger or a “one-size-fits-all” strategy that manage to deliver productivity gains to all firms alike. The impact of digitalisation on the average firm’s TFP growth is relatively small and we find that most firms are not able to benefit from digitalisation at all...intangible assets are closely linked with digital technologies and act as a complement to digitalisation to magnify its positive impact on the average firm’s TFP growth.”

Hence, the transition to a digital economy has been verified as a precedence by policymakers and international institutions and so implementing numerous policies in order to speed up digital adoption and in that way enhancing productivity.¹⁴

It could be said that corporations only in some areas benefit on average from investing in digital technologies, which means that digitalization is a game changer only for corporations implementing digital technologies to become considerably more productive over time. Hence, a higher digital

investment by itself is not turning highly productive “laggards” into frontier corporations, which means that corporation’s surface at the top of the productivity distribution is achieved via a more structural innovation strategy and not directly after an increase in digital investment. Also, it could be said that policies that incentivize corporations to become more digital over time should be constructed in a manner assisting “laggard” companies to fully reap the productivity returns from higher digital investments on a case-by-case basis, and so corporations must implement new digital technologies under unique conditions in their daily operations.¹⁵ To that extent, investment treaties grant protections and incentives to foreign investors, allowing them to pursue remedies if they consider these provisions have been breached.

It has to be taken into consideration that digitalization boosts the effectiveness of information collection, storage, and exchange, making possible the collection and processing of information in a fully automated way. Moreover, digitalization permits storing large amounts of detailed data and relies on a fully automated data exchange, and so, digitalization generates economies of scale and network consequences, shaping the economic basis for the speedy advancement of platforms. Hence, digitalization produces room for new products and services, such as e-books or personalized items, thus advancing not only the efficiency of inventory flows but also tracking and recognizing customers. Thus, digitalization generates the processing of information in a fully automated manner, such as analyzing data by AI, permitting a fully automated data exchange between market participants or members of the value chain, boosting the evolution of platforms. In addition, personalized advertising augments consumer choice by lowering advertising costs and permitting new entrants to market their products without a massive high marketing cost,¹⁶ which means that personalized ad-based monetization permits apps to advance “high-quality services to consumers for free.”¹⁷

It is worth noting that the new players are cloud operators in industries with a significant digital goods or services component, gaining deep insight into their cloud users’ needs and business models, creating another channel for insider trading. On the other hand, AAI digitalization will alter the whole concept of material life on earth grounded on the rationale of

introducing AAI humanoids consuming a particular kind of energy, which might alter the global production unless AAI humanoids have the same feeding demands as humans. This means that the laws of conventional economics will apply to the AAI economy as well, differentiating only in the tools of implementation.

It has to be taken into account that the consequences of digitalization on the manufacturing area, where digital means play a less direct role, vary considerably from their influence on traditional services or digital products, which could gain the most from digitalization and thus expose noteworthy disparities in how digital transformation remodels global trade.¹⁸

2.3 Centralization versus Decentralization

Centralization encompasses the central bureaucratic decision-making that most businesses undertake, and second a centralized ledger records all data, information, and transactions such as a corporation's assets, liabilities, costs, and revenue. From an organizational standpoint, centralization applies to government structures and at the core of corporations, which means that users have no control over the single decision-maker running centralized systems, resulting in possible censorship. The best possible level of decentralization necessitates balancing the costs of poor decisions owing to poor information and those arising from goal incongruence which means that deciding on the optimal level of decentralization implicates a trade-off between coordination and adaptation.

It is characteristic that the world is becoming centralized and so gradually is controlled by a few MNEs to the impairment of the many. On the other hand, decentralization offers more effective management in the financial sector, greater privacy protection, and fairer interfaces between the public sector, private sector, and people domain which means putting humans at the center of decision-making. Also, due to globalization and access to technologies, more and more interfaces between the public sector, the private sector, and people have become digitized, and so, digitalization connects people, businesses, and states empowering access to public goods, generating digital identity, and improving financial dealings via cyberspace and AI systems, which means that e-commerce and social media link people

everywhere. In other words, the capability to digitally interconnect and share information is enabled by cyberspace and AI space.

It is worth noting that cyberspace became centralized in many ways, and so, cyberspace supposed to be decentralized. In the end the wish of states to control information led to the centralization of cyberspace.¹⁹ Hence, following the centralization of cyberspace, there is a tendency toward digitized decentralization of finance and money, which in the end will be centralized via new technologies and controlled by a few corporations and governments as well. To that extent, the arrival of Bitcoin and the “decentralized finance” or DeFi establish a type of decentralization allowing the exchange of value without the risk of requiring to face up to a single intermediary. Despite that the Bitcoin system started as decentralized, it seems that the Bitcoin system has been displaying escalating signs of control and centralization.

A certain picture of centralization is money. Money, of the fiat kind, like bank notes and coins, is issued by the central bank of a country in a centralized manner and backed by and supported by that central bank and its centralized government. Moreover, in a centralized system, a single decision-maker or central authority is realized. To that extent, cyberspace has been centralized at the commercial and organizational level and so, digital platforms utilizing AI systems have become near monopolies in their given area of functionality and there has been a concentration of ownership in the organizers of connectivity rather than providers of specific functionality. In addition, hardware ownership and management are centralized by cloud providers along with the utility software embracing operating systems and databases.

Are governance and decision-making crucial elements in appraising whether a system is centralized or decentralized? A system with fully decentralized decision-making implies the absence of any form of influence, power, or control over developers or contributors. Taking into consideration that decentralization runs on and within a spectrum, a system is decentralized for the reason that there is no central decision-maker or authority; however, a decentralized system can be more or less decentralized than another system based on the fact that various variables and factors appear internally and externally within and around a system,

influencing the interaction with any other system on social, political, technological, and legal matters.

It has to be taken into account that the emergence of cryptocurrencies and the decentralized finance or DeFi are responses to centralization. Decentralized finance comprises widespread networks of open source, but not all systems are decentralized across all dimensions. However, they have altered the understanding of how digital systems operate, introducing new methods of authentication that are more automated than those of central governmental authorities, big banks, giant payment service providers, or social media corporations. Digital networks require no enrollment at all, and the databases that they utilize do not sit on that corporation's servers. There is a fast-changing and technologically driven world due to the fact that everything is digitized and is becoming smart through the use of data, which denotes more and more digital representations of things, like land titles, currency, commodities, and contracts that are recorded, exchanged, and stored digitally. Cloud computing is a centralized phenomenon, and thus, the Cloud is an example of centralization in action. It is characteristic that decentralized finance has not yet reached its full potential as a result of several challenges linked to fraud, volatility, usability, and regulatory uncertainty. Therefore, an AAI system controlling and correcting directly any deficiencies in the whole process will solve the ineffectiveness of present decentralized finance models.

It is argued that the capacity to transfer money across borders in a safe and secure manner is a prerequisite for the global economy. The prospective advantages of Central Bank Digital Currencies (CBDCs) in progressing financial stability, establishing financial inclusion, and boosting the competence and resilience of payments have been achieved owing to the declining use of physical cash, and so, CBDCs are turning out to be a geopolitically strategic priority and even confronting the supremacy of the US dollar.²⁰ Financial inclusion is the competence of people and corporations to have effective access to a wide range of appropriate financial services. Financial inclusion and access to credit are factors in accomplishing financial sustainability.

It is stipulated that central banks must reexamine their role as money issuers²¹ in the digital age. Thus, central bank digital cash is a way of

upholding financial stability and limiting the mass-privatization of money, which will generate problems of trust and the quantity of money circulating in the markets which means inevitably problems of inflation and real growth.

The responsibility to manage and preserve the value of currency is not the same as the responsibility to manage and administer transactions, and the responsibility to supervise payment systems is not the same as the responsibility to run them. Hence, the management of a transactional infrastructure supporting CBDC is not antagonistic to the operational responsibility of a central bank in using CBDCs to produce money and carry out monetary policy. The European Central Bank sees that CBDC could be an alternative to euro banknotes and could complement cash by serving as “an electronic means of payment that anyone could use in the euro area.”²² IMF regards CBDC as a new kind of money serving as legal tender. Having a more decentralized governance and running distributed ledger technologies (DLTs) allow participants to better compete and innovate and possibly offer better and more added value to end-users and customers. To that extent, Eddie Yue²³ highlights the “need a holistic digital money framework that will facilitate the healthy co-existence of public and private money to support our future digital economy.”

As the concept of centralization compares to the concentration of power and control, decentralized finance or DeFi is a new mode of operating financial services with the concept of decentralization at its heart. To that extent, the impetus for DeFi is to remove centralized middlemen, such as banks, brokers, or exchanges and in doing so DeFi is using blockchain technology to facilitate peer-to-peer (P2P) transfers, borrowing and lending, along with trading of digital assets.

To that extent, financial technologies have transformed the banking sector, and there is an escalating competition between banks and emerging fintech corporations due to the fact that fintech has reformed banking operations and services. In line with this, innovations offer a platform for banks to distribute their existing services faster and more competently because by using AI, banks offer chatbots and virtual assistants, establishing a direct connection with clients and thus allowing the delivery of multiple services such as investing in cryptocurrencies or online banking without the

requirement for human intervention. Likewise, fintech technologies allow non-bank corporations to deliver financial services conventionally offered only by banks, and so, new competitors will substitute traditional banks altogether.²⁴ It is characteristic that fintech's effect is in payments, clearing, and settlement services, and so, fintech assumes some of the functions that have conventionally been filled by financial institutions.²⁵

Moreover, DeFi generates and functions as decentralized financial service offerings and applications on open-source software platforms²⁶ and so, DeFi protocols wrap a wide selection of financial services, ranging from lending and asset management activities to decentralized exchange functionality, and processing derivatives and other financial transactions. In fact, platform lending has changed the way people engage in lending and borrowing around the globe. Additionally, online platform lending enhances the efficiency of financial intermediation, which is not always conceived of as one of the central functions of banks. Chris Clarke²⁷ argues that

”While platforms are likely to be involved in a degree of integration with the “installed base” of existing capitalist sociotechnical devices and social relations, there is also always the possibility of reimagining their design outside the narrow financialised outlook promoted by financial inclusion initiatives.”

Also, DeFi works as a decentralized financial system, entirely without intermediaries such as financial institutions. On the one hand, in centralized finance (CeFi), trust is generated by the reputation and guarantees of intermediaries. On the other hand, in DeFi, trust is produced by the technical properties of the underlying blockchain technology, such as cryptographic techniques and consensus algorithms generating transparency and immutability, not mentioning that DeFi's absence of intermediaries enhances working effectiveness and enables wider access to financial services. Furthermore, DeFi initiates a stable financial system that does not rely on centralized financial institutions or large banks.

DeFi embraces distributed ledger systems and smart contracts, presenting an alternative to the existing financial system. Hence, blockchain

technology, in conjunction with smart contracts, offers the technical basis for DeFi applications. In the traditional CeFi setting, the lender stereotypically assesses the level of risk of the borrower, involving extensive know your customer (KYC) and anti-money laundering (AML) verification as the basis for defining the terms and conditions of the loan agreement. In the DeFi space, the lender cannot appraise the level of risk since the borrower is naturally anonymous; thus, the lack of risk evaluation is characteristically made up for by overcollateralization, which means that this overcollateralization protects against the high volatility of crypto assets.

The blockchain technology underlying the DeFi protocols guarantees true immutability using cryptographic methods and consensus algorithms; thus, immutability merges transparency and trustlessness because manipulations are transparent to all, and the network rejects illegitimate changes. DeFi gives everyone permissionless access to financial products and services at any time. From an economic perspective, the central risks and disadvantages of DeFi related to CeFi correlate with oracle dependency, the design of stablecoins, high levels of leverage, regulatory uncertainty, and possibly high barriers to entry for less well-off users participating in lower-value transactions. Moreover, the risks associated with oracle dependencies arise from the fact that many DeFi protocols are grounded in information which is offered by third-party providers. Consequently, DeFi protocols have to rely on the reliability, accuracy, and correctness of this information, causing a security threat through malicious oracles that have an incentive to deliver incorrect information.

It is worth mentioning here that Bitcoin is seen as a speculative investment instead of an alternative currency or a medium of exchange even though several MNEs accept Bitcoin payments. It is characteristic that the problem with cryptocurrencies problem lies in their high volatility as a result of the inability to place a correct valuation, and so, if products and services are priced in Bitcoin, corporations will have high price-changing costs owing to the high volatility of Bitcoin.²⁸ To that extent, due to the high volatility of cryptocurrencies, stablecoins have been presented to the market and are anticipated to have stable value as they are pegged to another asset.²⁹ On the other hand, it is argued that the spillover effects of

Bitcoin bring volatility on stablecoins.³⁰ In line with this, B. Thanh³¹ et al. argue that

“the price movement of large-cap stablecoins can be interpreted as signals on trading opportunities for smaller-cap stablecoins by investors. Therefore, large-cap stablecoins tend to drive smaller ones. The findings are expected to show how larger stablecoin price changes influence the prices of lower market cap stable coins because stablecoins users can easily switch between different types of stablecoins in the cryptocurrency market”.

As mentioned earlier, DeFi creates a new form of financial intermediation, broadens financial inclusion, and augments operational efficiency stemming from the qualities of transparency, immutability, and trustlessness linked with the underlying blockchain technology, and from the character of the DeFi ecosystem to accelerate composability among highly interoperable protocols executing financial roles and services, such as lending, borrowing, and exchanges. Nonetheless, DeFi risks involve scalability issues of the underlying blockchain technology, privacy concerns owing to the transparency property of blockchains, and legal and regulatory reservations, such as the lack of KYC/AML checks.³²

2.4 Financial Globalization

It is characteristic that globalization refers to the increasing integration of national and international financial markets, as well as the increasing international presence of major banks and other financial intermediaries, and consolidation refers to the growing size and cross-sectoral scope of financial institutions. Consolidation and concentration in the financial services industry have been raised particularly in the origination of syndicated loans and in derivatives markets. It could be argued that interactions between the effects of consolidation and the problems arising from marketization and globalization give further grounds for more

extensive use of public safety net instruments. Recently, the global financial system has changed deeply, and technological improvements and deregulation have increased competition in the financial services industry. Hence, globalization has implications for the generalized injection of liquidity in the face of financial difficulties, inducing national monetary authorities to react to international as opposed to solely domestic objectives.

Financial globalization offers measures and inducements for corporate insiders to better protect the rights of their minority investors.³³ Supplementary, investment in corporate governance is less beneficial in states in which the agency problem of state ruler option is noteworthy for the reason that many actions that establish corporate insiders decrease the risk of expropriation by the state.

It has to be taken into account that corporate insiders' trading conduct and investment horizon differ because of differences in investment patterns, liquidity desires, information access, and fears concerning restrictions on insider trading. Moreover, insiders are subject to some trading incentives that are not applicable to other groups of investors, such as delegated asset managers. It is characteristic that the trading conduct of insiders is affected by the horizon of their compensation contracts and so, many insiders build up shareholdings through their compensation package. Also, insiders have reasons to use their privileged access to private information to exploit trading profits by timing their purchases and sales. To that extent, Ferhat Akbas³⁴ et al. argue

that “an insider’s investment horizon establishes a benchmark for expected patterns of continued trading behavior and thus helps identify unexpected insider trades, which should be more informative in efficient markets...the trades of short-horizon insiders are both more unexpected and more informed, on average, than those of long-horizon insiders”.

The significance of efficient financial markets lies in allowing those who want to obtain capital from different sources at reasonable rates. Financial intermediaries activate scarce savings, evaluate research projects, manage risk by evaluating future cash flow, and facilitate transactions. Moreover,

financial constraints arising from imperfections in financial markets are a result of both information asymmetry and the politico-institutional environment.³⁵

Enterprises globalizing their operations and the associated capital flows have become key players in the world economy, and cross-border conduct and capital flows span institutional settings with shifting investor protections and levels of capital market expansion. Does the rapid globalization of financial markets take humanity into uncharted territory? Global finance advances via new digital financial instruments. Is massive international lending a new phenomenon? The gold standard allowed huge flows of capital to search the world for the highest return, permitting rapid growth and fostering international trade, making gold standard look like the golden age of the international monetary system.³⁶ Capital mobility is today very high for short-term instruments but not for long-term capital movements.

It has to be taken into account that international capital mobility has prevailed in periods of political support either for subordinating monetary policy to exchange rate stability (as in the gold standard) or for giving up exchange rate stability so as to enable monetary policy to pursue domestic objectives (as in the post-Bretton Woods era). Besides, when countries attempted at the same time to target their exchange rates and use monetary policy in pursuit of domestic objectives, they had to impose controls to curtail capital movements. International currency, flexible exchange rates, and sound institutions are considered benchmarks that defines conditions under which countries can integrate effectively into the world financial markets. In contrast, weak currency, fear of floating, and weak institutions are experiences of developing countries as they integrate into international financial markets. Financial market interlinkages create extensive interdependencies in the course of which a shock in one part of the system can be transmitted to others, exacerbating the contagion of problems from one economic unit to another.³⁷

The globalization of international trade, in which markets are opened to flows of foreign goods and services, is necessary. To that extent, financial globalization has several crucial benefits in emerging market economies by bringing in new capital, lowering the cost of capital, and promoting

investment, which in turn encourages growth. Hence, when foreign capital and financial institutions enter a state, they enhance the allocation of capital. Moreover, financial globalization needs to be widespread enough so that the entry of foreign capital and foreign institutions boosts competition in home financial markets. In fact, AI financial globalization is not only a choice but also a need to be the focus of the next great AI globalization.³⁸ Credit money is the form of money mostly circulated under conditions of financial globalization, and thus, due to instability of credit money, general instability has emerged in financial markets globally. Can uncertainty be regulated? The answer is no; however, ex ante rules will alleviate the consequences because uncertainty, being the motivating factor not only in generating profits but also in the character of financial markets, cannot be removed by regulation from financial markets. To that extent, Chris Clarke³⁹ argues that “uncertainty in finance might be beyond regulation but at the same time there is still a societal need to attempt to achieve a politics of uncertainty that can cope with ignorance of the future through experimental governmental efforts.” On the other hand, AAI technology run by AAI humanoids and operating objectively within financial ecosystems will bring fair growth by correctly utilizing global resources.

Finance is subject to problems like incomplete markets, asymmetric information, and bubbles, requiring a necessity to prevent financial instability by controlling financial flows which means that the introduction of regulations within national boundaries, taking the form of preventing financial flows that are mainly geared toward speculative activity, is necessary. In other words, regulations must bring financial stability and promote governments’ economic and social objectives. The productive sectors must neither be constrained by the price of finance nor by inadequate financial flows.

Financial globalization will develop a global institution that can play the central coordinating role.⁴⁰ Zekos considers that, in the AI era, IMF should take on new responsibilities and contribute to the harmonious financial development in a digital and global world. It is argued that a single currency will allow the global financial market to develop uniform credit standard requirements and, in the absence of the latter, this process will cause periodic financial crises and further intensify the unequal distribution of

income and wealth, which immediately contravenes the true meaning of the word globalization.⁴¹ Also, financial development advances the effectiveness of capital allocation and lessens information and transaction costs, which means higher rates of technological growth, capital formation, and GDP growth.

To that extent, Zekos considers that the interplay of values among strong currencies is a dynamic factor for the movement of capital, creating development opportunities worldwide. Greenspan⁴² suggests that

“A clear benefit of financial globalization is that, to the extent that it reduces home bias, savings will be better directed to the most promising investments in the world, increasing global economic growth and prosperity. However, so long as risk aversion exists and trust is enhanced by local familiarity, we cannot expect that home bias will fully dissipate”.

Moreover, Zekos believes that financial globalization allows for a better environment where capital will be moved around the world with fewer barriers, where productivity will bring higher profits. Thus, financial globalization means growth for developing economies as global savings are resourcefully directed to capital accumulation in developing countries where the marginal product of capital is highest. Integration requires individual countries to have the capability to attract financial capital, which is independent of external barriers and is determined by the internal state of the economies of the countries concerned and the future opportunities these economies can offer.⁴³ Developed and newly formed countries undertake various measures to prevent financial instability by controlling financial flows that are purely speculative in nature, ensuring that possible expansion in aggregate expenditure in the productive aspect of the economy is not constrained by inadequacies in the financial flow.

It is characteristic that financial globalization has diverse consequences for bank risk-taking, market power, efficiency, trade, economic growth, and business models. In fact, banks generate liquidity by transferring liquidity liabilities to illiquid assets or by offering off-balance sheet activities such as loan commitments or standby letters of credit. In other words, bank

liquidity making assesses total bank output and shows a strong positive correlation between bank liquidity and real economic output, which means that liquidity making plays an imperative part in the economy.⁴⁴ Also, currently, AI financial globalization boosts liquidity construction by bringing additional lending resources to the banking systems of host countries. Subsequently, financial globalization and bank liquidity have implications for policymakers when articulating appropriate measures to deepen financial globalization.

Has AI financial globalization yield equitable benefits for all nations? In fact, foreign investors encounter informational barriers because of the gap between home and host countries and so, becoming cautious in taking risks, which means a reduction in liquidity creation. In line with this, S. M. Aamir Ali and A. Ghose⁴⁵ argue that

“Although the concept of globalisation was initially intended to enhance the agency of developing nations, it has instead resulted in the erosion of their autonomy to independently determine their course of action. The democratic process has been impacted by the loss of decision-making power experienced by emerging countries as a consequence of globalisation. The rules governing globalisation were inequitably formulated to provide preferential treatment to developed nations. Contrary to prevailing popular opinion, the impact of globalisation on both developing and affluent nations has exhibited disparities”.

The widespread deployment of robots and AI-enabled systems in many fields of economies globally has generated a digitized financial capitalism encompassing a shrinking workforce. In fact, digital capitalism is a transformation of the capitalist system of production in which digital technologies comprise the central production and control device in the global market which means that capitalism becomes digital when the production procedure of particular commodities is accomplished by and through privately owned digital technologies. Thus, digital capitalism

encompasses the processes through which digital technology facilitates the structural tendencies of capitalism.⁴⁶ In fact, digital capitalism by encompassing digital networks is becoming vital part of capital assets worldwide by embracing metadata collected through cyberspace platforms, generating the formation of new market types in the exchange of goods, services, and money. Also, digital capitalism necessitates new work categories due to the fact that digital technologies serve as labor devices and infrastructure and so having a key role in certain production methods, as digital media are today both productive technologies and methods, which means that digital capitalism generates new managerial approaches because digital technologies are utilized as managerial means within corporations. It has to be taken into account that the relationship between digital capitalism and the digital/AI economy is that digital platforms have a key role in permitting people to become independent contractors working as freelancers, while employers choose the individuals that best fit their requirements.⁴⁷

2.5 The Market, Banks, and Cash Management

It has to be taken into account that the market has risky assets, traded by informed investors, noise traders, and competitive market makers. Moreover, information transmits from the less informed investor to the more informed investor. Also, market competence advances or market liquidity degenerates according to the characteristics of the less informed investors' information, and so, total trading volume can be higher or lower.⁴⁸ To that extent, Itay Goldstein et al.⁴⁹ argue that

“the well-informed investor never shares information because doing so only dissipates his informational advantage and erodes his profit accordingly...an investor with coarse information optimally chooses to share information with a well-informed investor, because the former benefits from the latter trading against the shared information, which therefore dampens the former's price impact. By contrast, the well-

informed investor never shares in-formation because doing so only dissipates his informational advantage and erodes his profit accordingly”.

It is representative that interactions among investors are influencing trading strategies and financial market outcomes. Taking into account that sharing information is costly to investors due to the fact that it diminishes their informational gain, it is vital to understand the assistance investors get from sharing information.⁵⁰

Banks wield deposit market power by investing in capital and, by holding reserves, circumvent costly liquidity deficiencies but do not entirely internalize the benefits. Anyway, banks put their positions/titles to hedge funds as a result of the higher balance sheet costs inflicted on banks by regulatory alterations, such as the launch of the leverage ratio after the global financial crisis. It has to be taken into consideration that the central bank’s selection of deposit subsidies and interest on reserves influences the cost structure of banks, their price setting, and so the liquidity premium on deposits. More bank market power necessitates a higher deposit subsidy to inspire banks to lengthen their balance sheets. To that extent, Dirk Niepelt⁵¹ argues that

“unlike in the single-tier system, optimal policy in the two-tier system with its distortions requires fiscal resources or regulation to correct them. Dead weight losses of taxation or regulation therefore add to the social costs of the two-tier, but not the single-tier, payment system”.

It is well known that banks are fragile, encompassing the likelihood of individual banks suffering a run, as a result of the provision of liquidity transformation and ultimately interlocked through fire-sale spillovers. Moreover, when facing withdrawals, banks liquidate their assets in a common market, which means that they compel negative fire-sale externalities on one another. Nonetheless, reducing bank fragility is welfare-improving for the reason that runs are linked to real resource losses owing to fire sales, and so, stabilizing the financial sector is an objective for

the regulator which means that regulators have to maintain strong banks because the fragility of strong banks affects weak banks on the margin but not vice versa.⁵²

The banking industry is undergoing a digital revolution⁵³ and so FinTech corporations and BigTech platforms are involved in conventional banking businesses using their innovative information and automation technologies. Also, “open banking” empowers borrowers to disclose their customer data stored in a bank with a fintech that has AI information processing technology but less access to customer data. Moreover, open banking enhances the fintech’s screening competence, but, on the other hand, it diminishes lending competition and harms borrowers if the fintech corporation is “over-empowered” by the data sharing means.⁵⁴ Also, banks with higher IT adoption have larger loan growth due to the fact that an enhancement of a lender’s IT augments its lending volume.⁵⁵ Cyberspace and AI employment exaggerate bank competition, resulting in lower banks’ loan prices.⁵⁶ Also, companies’ lending by banks with higher IT adoption is less influenced by the distance between the bank headquarters and their borrowers. To that extent, Xavier Vives and Zhiqiang Ye⁵⁷ argue that “whether the development of information technology intensifies lender competition depends on its impact on differentiation.”

Reliability of the market is decisive for financial stability. Capital regulation is a path through which accounting rules influence bank behavior and financial stability. It is worth mentioning here that some lenders might be part of criminal organizations that use IML to launder money and conceal profits from other criminal activities due to the fact that borrowers, discarded by any legal creditor, invest IML loans encompassing gambling, drugs, and alcohol. Moreover, the existence of IML is boosted by the extensive global adoption of interest rate caps limiting access to legal credit for risky borrowers, which means forwarding claims for illegal lending, and so, illegal money lending is dominant across the globe.⁵⁸ To that extent, K. Leong⁵⁹ et al. argue that

“the crackdown was highly successful at lowering the payoffs of lenders and borrowers in the market, as well as lowering the total volume of loans disbursed. Removing borrowers

from the market, either through offering formal market alternatives, rehabilitation or education programs, also hurts lenders, particularly if they focus on medium-performing borrowers in terms of loan repayment time”.

It is worth noting that high leverage coupled with short-term funding when exploited to finance risky assets is very instable exposing banks to runs, which means that banks’ disclosures concerning exposures to risky assets and funding structures play an imperative part in banks’ stability. On the one hand, more wide-ranging disclosures offer relevant information regarding exposures to problematic assets and so diminishing uncertainty. On the other hand, more disclosures generate bank runs, initiating a panic among investors. To that extent, Jannis Bischof⁶⁰ et al. argue that “banks’ disclosures about relevant risk exposures as well as the accounting recognition of loan losses came late, relative to the evolution of the crisis and concurrent market expectations. Many banks apparently were reluctant to communicate and recognize their losses.”

Taking into account that a financial system encompasses interrelated fragilities, several elements influence financial stability by altering bank heterogeneity. Hence, regulatory policies limiting the dispersion in bank asset performances generate a damaging course for stability because when strong banks are forced to liquidate their assets, the asset market is harmed as a result of massive liquidations by weak banks.⁶¹ To that extent, I. Goldstein⁶² et al. argue that “a certain difference between banks’ resistance to a common set of stress scenarios is desirable...Regulators should take a holistic approach and combine different regulatory tools to maintain a certain level of heterogeneity in the financial system.”

It is characteristic that banks create value by lending and cash management, thus generating broad and persistent economic value. Moreover, cash management services amplify the value of a corporation’s reputational capital. It is well known that cash management services move the supply and demand for products in wholesale and retail markets. On the other hand, functioning without cash management augments both counterparty risk and administrative costs by forcing suppliers to transport, verify, and store cash which, otherwise, financial institutes direct. It has to

be taken into account that problems regarding cash management services generate a “cliente effect,” permitting them to sell at a price premium, which means that retail customers assign a lower marginal value to products sold at “dispensaries” without cash management.⁶³ In line with this, E. A. Berger and N. Seegert⁶⁴ argue that “businesses with cash management charge higher retail prices (8.3%), pay lower wholesale prices (7.3%), and have higher sales volume (19%). Together, these advantages create a 40% increase in profitability.”

Furthermore, cash management services, such as safe storage, verification, and guarantee of funds, delivered by financial institutions yield substantial economic value for corporations and in case that corporations lack these services, they have to get alternatives offering internal warehousing of cash, which incurs costs to transactions and so switching resources from core trade. On the other hand, without warehousing, corporations utilizing digital banking, cryptocurrencies, and fintech solutions bear the costs of developing infrastructure to perform these functions. In line with this, E. A. Berger and N. Seegert⁶⁵ argue that “the benefits of warehousing exist regardless of whether national banks, credit unions, or fintech provide these warehousing functions.”

It is characteristic that considerable value to investors arises not only from financial planning concerning portfolio construction but also from cash flow planning and benefits offered by plan optimization, bringing forth rewards from upgraded financial well-being. Corporations make investment, cash savings, cash payments, equity financing, and payout decisions to maximize their value. In addition, corporations hold more net financial assets for the reason that equity has become cheaper relative to debt. Taking into account that only tangible capital is pledgeable and the increase in intangible capital usage, corporations hold more cash.⁶⁶ Negative cash flows generate a stronger cash flow volatility impact on cash restrictions, causing the noticeable drop of corporation’s value. It is worth noting that cash flow timing patterns offer valuable insights into corporate fundamentals and corporations’ exposure to aggregate risks, embracing effects for asset prices⁶⁷ and real quantities. Nonetheless, financially constrained corporations exhibit a more noteworthy cash flow timing pattern. Hence, financially constrained corporations with higher year-end

cash collections gain a higher risk premium. There is a link between realized macroeconomic variables and asset prices.⁶⁸ In line with this, W. He⁶⁹ et al. argue that “Macroeconomic perceptions generate the cross-sectional financing, investment, and valuation differences, thus driving the *time variation* in these equity market anomalies.”

Financially constrained corporations have high financing requirements, and so financially constrained corporations with high equity financing demands collect cash payments⁷⁰ by year end to circumvent the high agency cost of year-end unpaid revenue. In line with this, W. Hu⁷¹ et al. argue that “year-end cash collection leads to more risk exposure and higher stock returns.” It has to be taken into consideration that riskier corporations, encompassing a higher link between cash flow and the aggregate shock, have higher financing demands, which means higher optimal savings. While high cash collection diminishes profitability, higher cash collection indicates greater information transparency and thus reduces the cost of accessing the financial market.⁷² To that extent, W. Hu⁷³ et al. argue that

“Immediate cash payments negatively affect profitability, but reduce equity financing costs by increasing information transparency. Financially constrained firms optimally collect more cash at year-end when firms’ performance attracts more attention and information transparency is more valuable. Such behavior further results in greater exposure to aggregate productivity and financial shocks”.

First of all, subjective prospects influence corporate behaviors such as investments and external financing, while the evidence of biased corporate returns and growth prospects shapes corporate-level investment and external financing. Moreover, positive expectations of aggregate productivity inspire optimism among both corporate managers and investors, causing an increase in external financing and investment, a fall in financing costs, an upsurge in contemporary equity prices, and thus cutting a subsequent return. It is characteristic that activities of financially constrained corporations are more influenced by misvaluation shocks or

behavioral biases such as overoptimism. Consequently, with an upward revision in prospects of aggregate productivity, financially constrained corporations issue a lot of actually low-quality debt. In the interim, managers could deliver overpriced equity and debt to utilize investor optimism, which means subsequent return reversals for financially constrained corporations. To that extent, W. He⁷⁴ et al. argue that

“following upward revisions in expectations of macroeconomic conditions, subsequent equity returns are relatively low for small, risky, unprofitable, low-quality, and financially distressed firms. On the other hand, following downward revisions in expectations of macroeconomic conditions, these categories of firms earn relatively high subsequent returns. Thus, revisions in expectations of macroeconomic conditions induce strong predictable time variations in anomalies based on these characteristics. In particular, favorable revisions in expectations of aggregate productivity predict stronger profitability, quality, and low-risk anomalies but weaker value, investment, and size anomalies”.

In fact, financial constraints augment the mispricing instigated by credit sentiment or extrapolation.⁷⁵ Also, macroeconomic movements and monetary policy actions have a greater influence on the contemporary fundamentals and returns of financially constrained corporations. Moreover, the investment and debt issuance of corporations with lower credit scores are vulnerable to credit conditions.

It seems that bank borrowers count on credit lines to manage cash flow shocks, and lenders charge borrowers for liquidity provision. Moreover, corporations functioning with old capital are riskier for the reason that costly technology adoption hampers their flexibility in progressing to the latest technology, making them more subject to technology frontier shocks. Also, corporations embrace the latest capital or keep running with the existing mode, which is turning out to be outdated and less valuable over

time, meaning that by undertaking the latest technology, a more competent capital is instated.⁷⁶ It has to be taken into account that gross investment is larger when the corporation's capital is inadequate, given that current capital cannot be utilized for the operation of the new technology. Furthermore, expensive new technology implementation hampers corporations' flexibility in updating their capital stock concerning technology, furthering the risk dispersion between technology-adopting corporations and non-adopting ones.

Corporations use cash management to foster reputational capital, permitting them to demand better terms in the wholesale market and advancing quality guarantees to consumers in the retail market. The Total Cash Management System lets MNE to produce positive cash flow, ensuring its operational use to cause future cash flows which means that the application of Total Cash Management supplies a platform for both present and long-term objectives in the international financial management of MNEs, and so a MNE has a prospective to get financial resources owing to the geography of its undertakings. Hence, MNEs have easy access to the devices of both local and global financial markets, and so MNEs have to choose the most effective financial resources and their sustainable management. To that extent, crypto assets provide a new way of financing MNEs.

MNEs have a high credit rating, allowing them to access cheaper resources and the availability of liquid assets and intensive cash flow let MNEs to diminish the cost of obtained capital, real sales evolution, and an upsurge of investment attractiveness, which means offering financial stability and opportunities for obtaining capital. Moreover, MNEs are amalgamated into the global financial system, and their finances are its structural elements, allowing leading global banks to offer conditions for the transparency of their activities and thus escalating the opportunities to obtain debt funds from credit organizations. Corporations and MNEs try to raise financing when banks cut lending.⁷⁷ On the other hand, there are shortcomings for MNEs due to currency, transaction, political, and economic risks affecting the decision regarding the selection of financial resources.⁷⁸

2.6 Venture Capital and AI

VCs have a decisive role in the economy as a result of their uneven contribution to corporate R&D and innovation. Moreover, VCs offer liquidity to founders, tempting them to become entrepreneurs.⁷⁹ In line with this, Michael Ewens⁸⁰ et al. argue that

“while a post-product increase in cash compensation makes founding a startup more attractive than a contract that pays a flat salary, low pre-product compensation can still leave high earning potential founders holding substantial non-diversifiable risk...variation in the expected time to product development is a key driver of the relative attractiveness of entrepreneurship across sectors for potential founders with attractive options outside of entrepreneurship and in sufficient personal wealth to finance consumption while waiting for a product to materialize”.

It is worth mentioning here that AI utilization across businesses is on the uptick; however, there is some concern regarding the use of AI, involving algorithmic aversion and fear of machine bias.⁸¹ Taking into account that an algorithm is as good as the data it works with, AI can identify and correct human bias, and multiple devices exist that identify and adjust for such.⁸²

VC evolved to fund promising startups that otherwise lacked the established presence required to safeguard business loans in exchange for an equity position. Regardless of the advances in AI and VC's focus on tech startups, VC corporations themselves have failed to employ data analytics and machine learning in their own decision-making, relying instead on copying the investments of other corporations. The use of AI by VC corporations advances investment outcomes.⁸³ Moreover, VC has shifted away from that of an advisor offering governance and oversight to startups, to that of a speculative hands-off financier. Even though originally VCs played active management roles in delivering both oversight and corporate governance, such is no longer the case, and so VC corporations are less

involved in developing and growing emergent businesses, capitulating to the impulses of their founders whose business strategies are grounded in undermining the competition to dominate the market.⁸⁴ To that extent, VC corporations merely copy the investments made by other corporations. Also, while the absence of due diligence results in unsuccessful investment decisions, the use of AI in investment decision-making offers better investment results.

Furthermore, AI can alter the entire VC landscape by opening the door for a greater number of successful startup openings, while concurrently expanding corporation investments which means that by integrating AI into the decision-making process, corporations can mitigate the influence of human bias and deliver more accurate decisions than those delivered by human experts. Thus, corporations that exploit objective, data-driven approaches to investment platforms will outperform those that do not, and so incorporating data analytics into VC investment decision-making will augment profit opportunities and expand portfolio diversification.

It is worth mentioning here that specialization leads to lower risk in cross-border investments from a venture capitalist's point of view, which means that with escalating deal-specificity, the support and value improvement of a foreign portfolio corporation are guaranteed in international markets as a result of the investor's specialization. Moreover, as venture capital corporations invest in new businesses functioning in ambitious markets, these investments are typified by high levels of asymmetric information and uncertainty. In line, investors encounter institutional, geographical, cultural, and legal distance from their own countries, which means that this inaccessibility rises higher information asymmetries, investment risk, and costs for the investor if the target business is located abroad, and so the investor allocates more resources to supervising and managing the business abroad.⁸⁵ Concerning venture capital, A. Nörthemann⁸⁶ argues that

“the likelihood of a cross-border deal increases with higher levels of industry specialization. Furthermore, this effect is moderated by determinants at the institutional and portfolio company levels, reflecting the degree of information

asymmetry between the venture capital firm and the portfolio company.”

A crucial factor in the appeal of venture capital (VC)-financed entrepreneurship for high-earning prospective founders is the anticipated time to create a startup’s initial product for the reason that founder-CEOs’ cash compensation rises significantly after product development, thus lessening the non-diversifiable risk that founders encounter at startup birth. Consequently, a startup’s projected time to develop its first product is a crucial basis for the appeal of VC-sponsored entrepreneurship. To that extent, Michael Ewens⁸⁷ et al. argue that

“the increases in CEOs’ cash compensation after a startup begins product development are large enough that non-diversifiable risk held by founder-CEOs falls substantially after achieving this milestone...a startup’s expected time to developing its first product is a key determinant of the attractiveness of VC-backed entrepreneurship”.

Hence, increases in CEOs’ cash compensation after a startup begins product development are large enough that non-diversifiable risk held by founder-CEOs falls substantially after achieving the product development breakthrough.

On the one hand, Felix von Meyerinck⁸⁸ et al. argue that

“directors’ involvement in forced CEO turnovers can be expected to be assessed by the market and to affect directors’ reputation...Directors interlocked to a forced turnover experience an economically large...directors are not penalized for involvement in a CEO turnover *per se* but for forced CEO turnovers that are related to governance failures by the board”.

On the other hand, sacking a CEO is seen as an indication of malfunction in the supervising and is harmful to a director’s status.

It has to be taken into consideration that conditional risk not only clarifies profitability and investment, but also supervising conditional risk is imperative for judging the economic significance of variances, appreciating market competence, and assessing the performance of asset managers. N.J. Gormsen and C.S. Jensen⁸⁹ argue that “Overall, the impact of conditional market risk on anomaly alphas is larger than the impact of unconditional market risk, suggesting that researchers and practitioners should seriously consider controlling for conditional risk whenever implementing the CAPM in the future.”

It is characteristic that venture capitalists with high social status in the national market invest abroad due to augmented national experience. Moreover, the positive development of a portfolio corporation is expected when the investor is accustomed to business patterns in the particular industry of investment. Additionally, industry-specific knowledge is vital in ascertaining good investment opportunities and adding value to corporations.

Also, corporation-level factors such as industry specialization or ties to a foreign country⁹⁰ play a part in investment decisions. Antonia Nörthemann⁹¹ argues that

“the probability of investing abroad increases with higher degrees of industry specialization even though international deals are often associated with higher risk. This relationship illustrates how the liability of foreignness has a varying influence on an investor’s investment decision depending on the investor’s specialization in the specific industry to which the deal is assigned.”

It is worth mentioning here that venture capitalists do not seek long-term progressive growth of the business where they invested, since this would not let an acceptable return on all the capital invested, which means that they demand well-timed and substantive results.⁹² It is worth noting here that the active portfolio optimization allows investors to gain access to stocks, bonds, and private equity (PE) funds, and so ex ante commitment

has large consequences for investors' portfolios and welfare.⁹³ To that extent, Elise Gourier⁹⁴ et al. argue that

“investors want to change their capital allocations when capital is called—most often to increase the allocation. Hence, investors are under allocated to PE and willing to pay a large premium to adjust the quantity committed upon capital call. A direct implication of PE under allocation is that the demand for changing the amount committed is asymmetric. Investors would rarely want to reduce their allocation to PE, but highly value the option to top-up their allocation when capital is called”.

Furthermore, venture capital and private equity have achieved importance in the market, and so venture funding for R&D has stimulated the establishment of several startups. To that extent, Shaun Wang⁹⁵ indicates that an augmented level of R&D input improves the corporation's value growth.

Financial liberalization is one of the most important aspects of globalization involving both short-term capital flows and FDI. Moreover, financial globalization is illustrated by the fact that trading in the global foreign exchange market has far outpaced the growth in international trade in goods and services. Additionally, foreign exchange is increasingly being treated as an asset or a commodity, not just as a means of payment, with a parallel increase in the use of derivatives in foreign exchange markets. Changes in global conditions, such as the decline in nominal interest rates, gave a spur to capital flows to emerging markets. Moreover, international trade is changing because of the introduction of new technologies, innovations, and geopolitical modifications, which means they are opening up new prospects for international business and trade expansion.⁹⁶ Taking into account that performance-targeting incentives upset corporate policies, competitive forces stabilize the strain to meet performance targets via alterations in other corporate policies, such as investments in capital and innovation.⁹⁷

2.7 AI and Financial Markets

Markets are powerful institutions run by law encompassing public and private regimes fashioning entitlements, enforcing market exchanges, and restricting expropriation, which means that distinctive types of markets exemplify fluctuating normative concepts of the market. To that extent, Hanoch Dagan⁹⁸ et al. argue that “appreciating the significance of law as the infrastructure of markets reveals that no particular market structure is inevitable. Instead, every market order is the result of a complex set of legal and political choices.” On the one hand, markets are the invention of human societies, and the form of any market is grounded in the legal rules guiding actors’ behavior in the market. On the other hand, beyond the law, markets function according to economic principles articulating their running. In other words, while markets are in part legal constructs shaping their construction, competent markets are invented so as to maximize economic output. To that extent, an AAI technology will be able to illustrate the interplay between legal and economic principles in valuing the role of the market at any moment in order to make the right predictions.

The appearance of international digital trading settings has made it possible for corporations to globalize operations on different trading platforms. Long-term financial planning takes a considerable amount of time in every corporation. The financial market has emerged as a place where debt, cash, and financial assets are exchanged; thus, the financial market is a system encompassing a set of financial transactions that are traded and embraces all organizations that act as financial intermediaries, directly or indirectly. Moreover, financial markets are incorporating a procedure under which financial assets are traded professionally, cost-effectively transferring financial resources from surplus sectors to sectors with medium and long-term financial deficits, manifesting several benefits for borrowers and investors. Also, financial markets handle securities, other than the monetary market, and informatics allow dealing from outside the market through brokerage firms spread across different states owing to the globalization of financial activity and the merger of capital markets. Additionally, financial markets deal with the management of funds as a key function in economic growth, having an impact on individual wealth, companies, consumers, and on economic flows.

Has AI caused financial market disruption? It is characteristic that the running of the financial markets is interconnected with business activities since one of the roles of banks and financial institutions is to offer access to the capital required by corporations and individuals. Consequently, financial markets are under the influencing powers of digitalization and “globalization.” Also, corporations are seeking financing outside the banking sector, while the transformation of the business environment in financial markets is ongoing. It is worth noting that the financial market disorder is associated with the aftereffects of digital disruption for businesses and individuals, resulting in the direct and indirect spread of digital disruption to finance as a role and industry. To that extent, the arrival of electronic platforms has empowered numerous new activities such as the financing of corporations.⁹⁹ Which is the role of AI in financial markets? AI will offer the data and predictions about the development and functioning of financial markets, allowing an objective approach, spotting any instances of corruption, and preventing crimes such as insider trading and money laundering.

It has to be taken into account that financial markets activate financial resources, directing them toward investment with the purpose of motivating financial operations and advancing their prospective returns. Revolutions in capital market technology have transpired at the level of the technologies and financial instruments available in the financial market, which have globalized in a manner that boosts the flow of capital and the global capital market integration. Also, financial markets’ effective and resourceful character is grounded in their capability to enlarge the size of the market by reinforcing their institutional and regulatory structure in order to exploit the benefits and drop the risks of financial liberalization, predominantly capital account liberalization and the integration of global financial markets.

As mentioned earlier, digitization is the process of transforming information from analogue to digital format, such as paper letters to email communications, or paper receipts to digital receipts, and so corporations keep better control of business activity and the capability of starting some basic automation. Moreover, startups emphasize the point that technology entails a solid business model or a disruptive transformation so as to engross all the prospective automation effectiveness and growth capabilities

leading to digitalization, where corporations align and digitally transform their business model. Hence, digitalization is altering people's ways of searching for and buying products and services due to the fact that digital platforms¹⁰⁰ are fundamentally transforming the people's interaction. Also, digitalization offers businesses new modes to grow, reach prospective customers, and expand into new markets globally. The use of Chatbots, WhatsApp, Instagram, and other digital platforms has become common in B2B¹⁰¹ relationships, and so there is a move in customer behavior, a transformation in technology, and an alteration in the market. Hence, corporations must keep up and learn about their customers' emerging habits. To this extent, the digital platforms have started a global rivalry, which means that if one firm fails to meet expectations, the customer finds other routes.

It is characteristic that corporations optimize their current business models in order to build up new products, services, and so achieving efficiency and flexibility. It has to be seen that optimizing short-term profitability does not automatically guarantee optimum shareholder returns in the long run. Consequently, data analytics, cognitive systems, AI, data-based business models, and platforms are used to stimulate growth.

It has to be taken into account that in order to be valuable, financial control systems have to mirror a corporation's strategies and objectives. Management control systems should encourage managers and other employees to use maximum effort through a sort of monetary and nonmonetary rewards.

2.8 Metaverse and Non-Fungible Tokens (NFTs)

It has to be taken into account that the integration of AI with VR and AR technologies is amplifying their collective effects within the metaverse by facilitating the formation of hyper-realistic simulations, the development of AI entities along with the enablement of deeper, more expressively involving interactions between users and AI entities.

Additionally, it has to be taken into consideration that digital entities are reforming the conception of identity, escalating it beyond physical individuals due to the fact that AI entities are simulating human behavior,

presenting diverging levels of interaction, autonomy, and behavioral complexity within the metaverse, which means that AI humanoids can appear as virtual assistants, companions, or social media influencers¹⁰² interacting with people and organizations and guiding human behavior.

It is characteristic that the emergence of the Metaverse and NFTs has initiated substantial alterations in ownership and participation in virtual economies online. The Metaverse is a shared online space that combines elements of the physical world, digital spaces, and cyberspace in generating a unified, interactive environment. The metaverse reshapes digital and physical lives. It could be said that virtual reality, augmented reality, blockchain, and AI are central technologies in creating engaging experiences and making sure new content is both safe and dynamic.

It is worth noting here that “Metaverse Identity”¹⁰³ incorporates personal characteristics and economic essentials in demonstrating a new frontier in inspiring self-expression by enclosing technological advancements, clarifying an energetic interplay between innovation and identity formation.¹⁰⁴ It has to be taken into account that the perception of digital identities and management within the metaverse allowing interoperability is the basis of an interconnected metaverse¹⁰⁵ system. Hence, the standardization of metaverse identities allows the advancement of the metaverse ecosystem. Thus, the absence of standardization and trust across platforms diminishes users’ ability to keep social value across multiple platforms.

NFTs are unique digital assets. The rise of NFTs has provided creators, including artists, musicians, and digital designers who create the original content that becomes tokenized as NFTs, with new ways to monetize their work, permitting them to claim ownership and receive compensation in manners that were not formerly conceivable. NFTs, cryptocurrencies, and Web3 technologies are producing new paradigms for money, possessions, and markets.¹⁰⁶

Due to the metaverse’s qualities, there is a need for robust mechanisms to protect user data, guaranteeing safe interactions in digital environments. The underlying technology of NFTs is assembled on blockchain, a distributed ledger that documents transactions across a network of computers, safeguarding data integrity and security. NFTs’ transformative

prospective in digital ownership and asset management within and beyond the metaverse necessitates a move toward more intelligent, responsive virtual environments. The Spatial Web merges the physical world with virtual spaces, making the web more multidimensional.¹⁰⁷

Furthermore, NFTs assign distinctive identifiers and metadata to digital assets and so making certain their indivisibility and distinctiveness, which means that NFTs brought a transferal in the monetization of digital content, empowering creators to energetically assert intellectual property rights. Hence, via the utilization of smart contracts, NFTs not only enable direct monetization but also allow the buildup of royalties from secondary market transactions and so advancing economic empowerment among creators within both digital and virtual economies. Thus, NFTs augment the liquidity and utility of virtual goods, playing an influential role in the proliferation of DeFi and accelerating the convergence of virtual and real-world economies.

The liquidity challenges arise due to an unplanned need for cash or securities and so capital regulations oblige banks to have liquidity buffers, which contain buffers for intraday liquidity. Managing liquidity is an existential activity for financial institutions, and liquidity management is muddled by “legacy” payment arrangements and financial market infrastructures.¹⁰⁸

NFTs leveraged blockchain technology to offer a solution to the digital replication challenge, enabling the creation of unique, verifiable digital assets. The extension of NFTs to virtual real estate reflects a broader application of this technology, illustrating its capacity to facilitate ownership in entirely digital environments, notably within virtual worlds or the metaverse. Virtual real estate transactions via NFTs underline the economic prospects of virtual spaces, offering opportunities for investment, development, and commerce similar to physical real estate but within digital landscapes. Taking into account that AAI humanoids might be able to live in both conventional and digital landscape and so giving another dimension to the role of fiat and digital currencies due to the different demands of the two different dimensions of economic and material life of the inhabitants of both environments.

It has to be taken into account that digital platforms present the technical infrastructure needed for issuing NFTs, which means offering means that

convert digital content into blockchain-based assets with verifiable ownership. Therefore, digital platforms offer the space to act between creators and the broader market, which means managing effectively digital assets. Interoperability and standards are central to the functionality and prevalent adoption of NFTs across several blockchain environments, and so allowing NFTs to be effortlessly traded, viewed, and used beyond the confines of their original platforms. Moreover, interoperability lets NFTs formed on one blockchain to interact with applications and users on another.

It is worth noting here that the decentralized quality of blockchain,¹⁰⁹ while advancing advantages in transparency and security, also confuses the control and protection of personal information. Hence, although blockchain technology offers a level of security through its immutable ledger, the interfaces and platforms users interact with denote features of vulnerability, which means that possible phishing attacks, smart contract exploits, and other forms of cyber threats present real risks to NFT owners. Furthermore, due to that fact that NFTs run on blockchain technology necessitating substantial computational power, the energy consumption linked with producing and trading NFTs has caused concerns regarding their sustainability.

It has to be taken into account that the development of virtual worlds¹¹⁰ is producing a new virtual economy encompassing many different virtual assets of core value. At the same time, the composability of NFTs¹¹¹ with DeFi is generating new investment prospects in an open financial system. Also, the decentralized character of NFTs and the absence of a standardized technique to assess their valuations are generating exploitation opportunities for AML, and thus NFTs could be leveraged by illicit players in concealing the real value of financial transactions or other business activities. Therefore, unregulated NFTs can be utilized to implement financial crimes such as money laundering and for conducting other unlawful activities.¹¹² In banking, the use of AI in AML and fraud detection has altered financial crime inspections, leading to more targeted identification of financial crimes.

Taking into account that NFTs are unique goods conveying information about the source in commercial trade, Andrea Mccollum¹¹³ argues that

“NFT tokens could be considered mere ‘ownership receipts’ for creative works rather than digital goods that are sold using association with a brand.” Moreover, NFTs are digital tokens that are stored using blockchain technology and contain metadata tying the token to a specific asset. Hence, NFTs offer the purchaser a digital link to a linked asset stored using blockchain technology. Thus, the transfer of assets using NFTs serves as a device for monetizing all types of digital content. In line, Andrea Mccollum¹¹⁴ argues that “NFTs and other blockchain-based assets will likely impact many areas of daily life in the future. Brand owners should be afforded the same protection in these spaces that they enjoy for tangible goods produced in the real world.”

It is characteristic that blockchain technologies and cryptocurrency will not be treated differently from other types of commercial products under trademark law, and sales of NFTs amount to engaging in commerce.¹¹⁵ It is argued that trademarks linked with physical items offer protection for digital reproductions of those items, which means that the MetaBirkin NFTs “could constitute a form of artistic expression,” and selling the images did not make them simple commodities rather than expressive works¹¹⁶ and so, the court held that the “title ‘MetaBirkins’ should be understood to refer to both the NFT and the digital image with which it is associated.”¹¹⁷

2.9 Security and Tokens

DLT and the blockchain empower the digitization of any asset class as tokens and are shaping the manner toward future of financial markets. Security token offerings (STOs) are advancing funding, where digital tokens are issued as regulated investment products on the blockchain. Once the token offering has taken place, the tokens are traded on the secondary market. Does the uncontrolled offering of digital money without any examination of its value generate over-offering of money leading to devaluation?

Security tokens¹¹⁸ denote shares of ownership in corporate equity, fixed income, investment funds, commodities, or less liquid asset classes such as real estate or fine art, comprising a new form of venture financing having

several advantages because corporations easily reach a large investor base while dropping transaction costs.

First of all, transparency concerning the transactions is accomplished through the blockchain, and the interoperability of the blockchain solves the problem of lack of compatibility between different systems or databases, allowing self-custody of any tokenized asset on one platform.¹¹⁹ Another major advantage of STOs is the prospective liquidity offered through the likelihood to transfer and trade tokens on secondary markets, which means that security tokens merge the advantages of the underlying technology with the legal protection of conventional securities.

Security token offerings are a method for corporations to increase capital where they issue digital tokens as regulated investment products on the blockchain. By transferring a digital token, the participants transfer the ownership or other property rights over the asset represented by the token without the intervention of traditional intermediaries. J. Kreppmeier and R. Laschinger¹²⁰ argue that

“companies that intend to raise funding via STOs would be well advised to offer a pre-sale phase in their STO and assure their intentions to trade the tokens on the secondary market while already devising a plan for successful future trading. From an investor’s perspective, these signals can be interpreted as positive quality signals on the basis of which appropriate investment decisions are conducted”.

STOs allow investment in securities recorded on a blockchain. In line, Thomas Lambert et al.¹²¹ argue that “entrepreneurs should carefully design their security tokens and run their offering professionally because doing so may not just help them raise more capital, but it may also foster general awareness of the benefits of a well-developed STO market.”

2.10 Crypto Assets

A crypto asset is a digital asset that functions through the digitalized distributed ledger or the blockchain. Crypto assets have appeared as a key movement in digital transformation and, by utilizing new technologies, are distinguished as one method of advancing financial operations. Moreover, crypto assets are being integrated into financial services applications to make possible near real-time transactions, accurate data recording, and more efficient payment processes. Hence, crypto assets create efficiency advantages in the financial sector by transforming the performance of financial markets to enable the exchange of value without the necessity for a central authority or intermediary, which means that crypto assets generate openings for the financial sector.

It could be said that crypto-assets markets are fast advancing and reaching a point where they will epitomize a threat to global financial stability as a result of their scale, structural vulnerabilities, and augmenting interconnectedness with the conventional financial system. Also, the swift evolution and global quality of these markets create the prospect for regulatory gaps, fragmentation, or arbitrage.

The emergence of Bitcoin and other cryptocurrencies has generated an anonymous alternative financial system, which means that criminal networks now have new openings to circumvent detection and seizure of funds. Nonetheless, crypto is hardly ever used as a medium of exchange to purchase goods and services and so is normally converted from and back to fiat currencies, such as US dollars, which means that the entry and exit points into the crypto ecosystem are crypto exchanges adhering to international laws designed to mitigate illicit financial flows. Also, while cryptocurrencies are aimed to be anonymous, the environment in which the transactions clear on the blockchain presents a ledger that tracks the movement of funds; thus, blockchain transactions are quasi-anonymous which means that, via algorithms, it is clarified where funds enter, move through, are exchanged into different cryptocurrencies, and exit the crypto ecosystem.¹²² Thus, cryptocurrencies are another cost-effective network for moving illicit funds across borders. To that extent, John M. Griffin and Kevin Mei¹²³ indicate that “the network of scammer nodes is highly connected, likely indicating that there exist widely-used services that funnel funds for extremely large and possibly related criminal networks.” It seems

that clustering AAI algorithms can be used to detect Bitcoin transactions for moving funds across sovereignties.

It has to be taken into account that while public equities trade on established secondary markets like the NYSE or NASDAQ, crypto assets trade on hundreds of upstart markets.¹²⁴ Moreover, crypto networks and tokens are devised by developers and then, by using blockchain technology and often smart contracts, conceived to run autonomously after setup. It also has to be taken into consideration that cryptocurrency investors rely on networks and media for evaluating new information regarding AI and AI tokens, leading to unusual price outcomes. ChatGPT is utilized in the crypto assets market for predictions, and so, AI tokens revealed higher returns linked to non-AI tokens around the launch of ChatGPT.¹²⁵ To that extent, Ante, L. and Demir, E.¹²⁶ specify that

“crypto asset markets are highly fragmented. While there was virtually no relevant attention for AI tokens before the launch of ChatGPT...investors should closely follow the developments related to specific themed tokens as their dynamics can be different from the rest of the cryptocurrency market.”

In line with this, Jang Ho Kim¹²⁷ argues that “asset class recommendations by ChatGPT improves portfolio efficiency compared to more diversified portfolios...Interestingly, the three indicators chosen in our analysis are market indicators that are also published with daily frequency (e.g., breakeven inflation rate measures daily expectations).”

In the EU, the Crypto-Assets Regulation (MiCA) aims to restrict the issuance of stablecoins to banks and so-called e-money institutions. Moreover, the Regulation of Digital Operational Resilience Act (DORA¹²⁸) strengthening cybersecurity, and Regulation (EU) 2022/858, which regulates distributed ledger technology,¹²⁹ accelerate digital innovation in the interest of consumers and market efficiency. Only public authorities’ ex ante review, coordinated by ESMA, affirms a harmonized application of MiCA. Also, MiCA identifies its scope by referring generically to tokens

that are not yet regulated by existing EU financial legislation. To that extent, D.A. Zetzsche et al.¹³⁰ argue that “all crypto-assets could be deemed, at first and prior to any involvement of supervisory authorities, ‘transferable securities’ unless (and until) they are exempted and requalified by NCAs applying the classification criteria promulgated by the (Joint) ESAs under MiCA.”

It has to be taken into account that the “cryptoization” of finance refers to when crypto “assets are substituted for domestic currency and assets, and circumvent exchange and capital control restrictions,” demanding global regulation due to its digital nature.¹³¹ To that extent, Gordon Y. Liao¹³² et al. argue “that payment stablecoins are proportionally less employed for trading and leveraged speculation activities than the fiat counterparts.” Moreover, crypto tokens have been judicially regarded as a form of property.¹³³

It is underlined that decentralized financial technologies could be used to circumvent regulation via anonymization, while raising fears concerning regulatory enforcement and increasing jurisdictional uncertainty because of these technologies’ operations regardless of borders.¹³⁴ It is worth noting that decentralized financial services expand financial inclusion, enable open access, and generate new openings for financiers and innovators, and so, decentralized networks cut transaction costs and produce network effects without encountering monopoly costs. In other words, once a decentralized peer-to-peer network tries to dominate, it cannot accumulate enough monopoly power to monopolize the network and prohibit others from participating, letting everybody to gain from the network consequences and so broadening transaction opportunities.¹³⁵ Moreover, decentralized finance generates a trust-minimized and automated version of traditional financial infrastructure.

It has to be taken into consideration that data “proptertization” empowers data corporations to enforce monopoly power across the market by increasing barriers for prospective market entrants.¹³⁶ Monopoly power causes discriminatory trades in data collection markets as a result of information asymmetries and bargaining power disparities. To that extent, Z. Chadambuka¹³⁷ argues that

“Data ownership would raise what is already a high cost to democratic politics from data. Applied to data, private property would make a sharp switch from providing for political liberty through ensuring decentralization of power and head off down the path to supporting centralized political control in the data-powered internet age. It would also move the dial backwards, taking society out of contract and back to status...law should provide protection against dangers from not just data collection, but also data processing, trading, and use”.

Digital platforms that run on blockchain technologies offering products and services use their own digital currency to settle transactions and rely on decentralized governance. It has to be taken into consideration that the FTX failure was a classic liquidity crisis that turned into a solvency crisis, like that of Lehmann Brothers in 2008, and so, the crashes of Babel Finance, Celsius Network, BlockFi, and Genesis were more of crypto lending corporations. Moreover, each collapse demonstrates the same pattern of noteworthy interconnected centralized crypto intermediaries becoming unstable as a result of mismanagement, malfeasance, fraud, theft, and a general lack of transparency.¹³⁸ To that extent, Douglas W. Arner et al.¹³⁹ argue that

“while crypto was presented to the world as a new type of finance avoiding the risks associated with traditional finance, in practice, agency, operational and financial risks have evolved across the ecosystem and crypto has proven not to be immune from traditional risks of finance”.

Also, T. Yuyama et al.¹⁴⁰ specify that

“although the stock market is correlated with the crypto market, the correlation is negatively affected by crypto-

specific risks. Therefore, the contagion effects of a crypto meltdown on the traditional market are not yet apparent and limited. In fact, we find: extreme volatility in the stock market can cause extreme volatility in cryptos, but not the other way around. The finding is consistent with the case of FTX's collapse...stock-specific risks spill over to crypto-assets, and consequently increasing the linkage between both markets, while the crypto-specific risks have limited spillover effects at the current stage, and are contributing toward a lower linkage between both markets. Risk spillover is likely to be one-sided and limited from crypto-assets up to this point. However, this may be because traditional financial institutions have little crypto-asset exposure”.

Thus, if transactions between the crypto-assets industry and the traditional financial sector increase in number and size in the future, the connectedness through crypto-assets holdings may lead to risk spillover. To that extent, the risk in the crypto ecosystem on financial stability depends critically on how interconnected it is with the traditional financial sector. Likewise, Azar, Pablo D, et al.¹⁴¹ argue that

“financial stability risks are not extensive because the digital asset ecosystem does not provide significant financial services and its interconnections with the traditional financial system are limited. At present, the potential spillovers from runs on stablecoins that are backed by money market instruments represent the most salient financial stability risk.”

It is characteristic that crypto assets growth has been accompanied by major volatility, with most crypto coins going through several cycles of rapid growth followed by spectacular collapses. Hence, crypto asset markets act both as a source of shocks and as amplifiers of overall market volatility; thus, they have implications for financial stability because crypto assets

incorporate technological attributes and features, serving as means of payment by decentralizing finance,¹⁴² to store value, asset transfer, privacy, digital identity, and governance.¹⁴³ To that extent, Roshan Iyer and Adina Popescu¹⁴⁴ indicate that

“the interconnections between crypto assets and financial assets are lower compared to within their respective asset classes, in terms of both sending and receiving returns and volatility spillovers. Furthermore, we show that crypto assets primarily transmit spillovers to financial markets, although during periods of financial sector stress, the reverse may also occur.”

Some market participants see existing stablecoins as substitutes for fiat currency in the crypto-asset ecosystem, but they are not widely used outside crypto-asset markets.¹⁴⁵ Data gaps and challenges avert authorities from making more conclusive or empirical assessments of stablecoin activities. It is characteristic that crypto asset markets risk being subject to fraud by crypto asset sponsors.¹⁴⁶ Moreover, asset digitization through distributed ledger technology has transformed trading markets, and hundreds of crypto assets are traded on various crypto exchanges, which means that the group of tradable assets is growing.

Due to the fact that cryptocurrencies are highly volatile assets, it is critical for investors to form a comprehensive investment strategy by employing risk-management measures before entering this market so as to mitigate the prospective for market crashes. Thus, investors should embrace a comprehensive investment strategy by incorporating “trend following” rules and respecting the impact of transaction costs. In this context, Trinh Le and Ummul Ruthbah¹⁴⁷ argue that

“both individual and institutional investors can benefit from applying a momentum (risk overlay) strategy to invest in crypto assets, as it allows them to capture the upside potential and avoid the downside risk of these volatile instruments...”

investors should carefully consider the trade-off between transaction costs and portfolio rebalancing when implementing trend-following strategies in crypto markets”.

It could be argued that there is a need to estimate the value of crypto assets.

It has to be taken into account that crypto lending platforms contribute to the transformation by drawing crypto-asset deposits from retail customers with high interest rates and using those funds to make loans. Hence, lending platforms use customer assets to make undercollateralized loans, which means that when lending platforms use depositors’ assets to raise leveraged positions, and so, they escalate the fragility of the crypto ecosystem. Thus, lending platforms provide collateralized crypto-asset loans to retail customers and uncollateralized or partially collateralized loans to crypto-native institutions.¹⁴⁸ It is characteristic that Ruchi Patel¹⁴⁹ indicates that

“crypto-colonization has relied heavily on the neoliberal policies, populist discourse...The resulting coalition of the crypto state and crypto elite has facilitated novel forms of territorial, resource, and political control at local and national scales. Despite the novelty of the crypto political economy, it inevitably reproduces the uneven, extractive, and neocolonial dynamics of neoliberal development in that it feeds on Global South geographies to export value North. Bitcoin thus serves foreign capital investment, accumulation, and imaginaries rather than the needs and systemic challenges of the poor. What is most worrisome is that the new crypto order operates with little transparency, oversight, or accountability of potentially harmful practices—from the so-called aid programs of Bitcoin Beach to wider crypto development and land grabbing schemes throughout the coast”.

The economic value of distributed ledger technologies (DLTs) is growing due to their implications for the future of payment systems. The advent of

stablecoins has accelerated the development of central bank digital currencies. Is there a shift in the usage of stablecoins? Stablecoins started out as on-chain cash balances for crypto exchanges and traders, but they resemble a form of general-purpose money tokenized on public blockchains. It has to be taken into account that the business of crypto exchanges has been shifting toward institutional customers, as asset managers have adopted cryptocurrencies. Gordon Y. Liao¹⁵⁰ et al. reveal that,

“since 2021, each dollar of payment stablecoin has supported approximately half the trading volume in cryptocurrency assets compared to each dollar of M2 money supply supporting traditional asset trading...the correlation between the growth of payment stablecoins in circulation and crypto asset returns is relatively weak, unlike the correlation found between trading stablecoins and crypto asset returns as well as the relationship between M2 money supply and equity market returns”.

Hence, the rise of cryptocurrencies and decentralized finance raises central questions regarding the nature of trust and money in the digital area. Moreover, Raphael Auer¹⁵¹ et al. argue that

“the footprint of the current generation of cryptocurrencies does not mark a sharp departure from the existing financial ecosystem. Instead of relying on a trust-free—ie on-chain—environment, a new set of agents has come to the fore that is offering convenience, market access, transaction scale and liquidity to these markets in much the same manner as in commercial banking and securities trading, albeit without the same degree of regulatory and supervisory oversight”.

Additionally, Mosioma and Walker¹⁵² explain that, of the 16 leading cryptocurrency platforms, just four are subject to trading regulation. On the other hand, cryptocurrencies and their intermediaries, involving crypto exchanges, should be subject to the same kinds of regulation and oversight as economically equivalent asset classes and institutions, concerning financial stability, consumer protection, and AML/CFT standards.

Furthermore, it has to be taken into consideration that the degree of decentralization of DeFi platforms differs within the DeFi industry, and so, some intermediaries, known as systemically important crypto intermediaries (SICIs), have secured large market shares or offer critical valuation services to many DeFi Stacks. On the other hand, it is supposed that decentralized platforms have appeared in order to avoid intermediaries via blockchain technologies. To that extent, the malfunction of the Terra-Luna stablecoin algorithm initiated a large-scale decline causing severe volatility of crypto-asset prices and so, the crypto industry suffered losses.

It is characteristic that in the cryptocurrency market materialized pump-and-dump schemes, a traditional form of securities fraud, and so, fraudsters use social media platforms such as Telegram, Discord, and Twitter to first spread false information and manipulate the price of low market-cap crypto assets, and afterward they sell their holdings to make a profit, triggering a fall in prices and investors to lose money.¹⁵³ In other words, it seems that “pump-and-dump” schemes are fraudulent practices encompassing the artificial inflation of stock prices through deceptive statements, and so, these schemes involve selling overvalued securities for a profit, leaving unsuspecting buyers with valueless investments. Moreover, “pump-and-dump” manipulation becomes profitable in the presence of technical traders, who manipulate information asymmetry and produce returns by raising prices and selling immediately. To that extent, stock prices increase as a result of information asymmetry and the influence of spammers. Dean Fantazzini and Yufeng Xiao¹⁵⁴ argue that detecting pump-and-dumps in real time implicates high-dimensional data, which means that AAI technology is the answer to any “pump-and-dump” scheme by detecting the anomalies via the speedy analysis of data and so prohibiting any insider trading from taking place.

It is worth noting that crypto asset trading is subject to laws combating money laundering, illegal imports, terrorist financing, and tax evasion; thus, leading digital platform operators in the United States and abroad have acceded to regulation as money transmission services.¹⁵⁵ John M. Griffin and Kevin Mei¹⁵⁶ reveal that

“the “legitimate” crypto space commonly serves as the entry and exit point to the illegitimate space and in so is facilitating the cheap and easy flow of funds that is the lifeblood enabling both pig butchering and modern-day slavery. Overall, our findings suggest that criminal networks are moving substantial funds cheaply and without much fear of detection”.

It is characteristic that Bittrex, Coinbase, and Binance, among the leading crypto exchanges in the world, found themselves charged with functioning as unregistered securities intermediaries in the United States, and the SEC assumed jurisdiction over a broad range of crypto asset platforms under the Securities Exchange Act of 1934, which means regulating crypto asset platforms and public trading in crypto assets.¹⁵⁷ The extensibility of digital ledger technology gives crypto asset platforms the potential to disrupt financial services. Moreover, crypto asset trading creates risks to the financial stability of the broader marketplace, which means that there is a need for proactively neutralizing new technologies’ possibly destabilizing systemic consequences. Otherwise, there will be a need of backing up an uncontrollable and indefensibly self-referential financial system.¹⁵⁸

2.11 Crypto Market Developments

It is characteristic that progress in digital technology for financial services has led to growth in markets for digital assets, with overwhelming effects on the protection of consumers, investors, and businesses, embracing data privacy and security, financial stability, systemic risk, and digital tokens. Moreover, the likelihood of switching from traditional markets to the crypto market has augmented systemic risk; thus, informed investors begin increasing investments in the market, trapping uninformed investors into

the market and then dropping the price by cutting off demand. In other words, the crypto market functions as a shadow system of traditional markets, and so not regulating the crypto market means that the risk will merely travel from the regulated market to the unregulated one.

Originally, governments embraced a laissez-faire approach, leaving the crypto market unregulated and thus having to deal with the consequences of unregulated decentralized finance (DeFi), which is based on cryptocurrencies.¹⁵⁹ H. Y. Jabotinsky and R. Sarel¹⁶⁰ argue that “empirical analysis reveals an interesting pattern: investors initially shifted funds to the crypto market when the pandemic erupted, but then made a U-turn and diverted funds out of cryptocurrencies, leading to a plunge in the market.”

For the reason that many cryptocurrencies behave as securities, the SEC began enforcing the usual securities regulations regarding disclosure on some existing tokens.¹⁶¹ An incorrect interpretation of a token’s current price is possible if investors rely on incomplete or false data, and so, herding increases systemic risk, which means that price bubbles arise and, simultaneously, bubbles are bursting when people herd in the opposite direction.¹⁶² Thus, herding in the crypto market is expected as long as much of the relevant information required for correct pricing is dispersed. It has to be taken into consideration that the crypto market will either lessen systemic risk by granting investors an alternative during a crisis or exaggerate the problem through spillovers to other markets.

AI agents, by being autonomous software entities capable of adjusting, learning, and accomplishing multi-step functions, are restructuring economic formations and governance in decentralized settings. In addition, AI agents are altering how machines interact with their environments and execute missions autonomously, and so, their undertakings are extending from simple rule-based systems to high-tech entities powered by LLMs. To that extent, the current evolution in LLMs has inspired the design and functionality of AI agents.¹⁶³ The market for AI agents in the crypto economy is just arising; therefore, monetary values and monetary policy of states will be tested first in a global environment and later on the AAI society will give another substance to monetary value in a purely AI economy.

It is characteristic that within GOAT's introduction, the token's market capitalization rocketed and the spread of Terminal of Truths (ToT) not only epitomizes the power of AI agents to affect markets but also signals a quickly expanding "AI agent meta" within the crypto ecosystem and digital markets.¹⁶⁴ To that extent, Ante, Lennart¹⁶⁵ refers to the fact that "in October 2024, an AI agent known as Terminal of Truths (ToT) stunned cryptocurrency markets by becoming the first such entity to achieve millionaire status through its holdings of the meme coin GOAT."

On the one hand, the prospect for abuse by AI agents offering investment advice generates matters of insider trading. On the other hand, due to the fact that AI and blockchain technologies converge, there is a demand for an effective, transparent, and equitable financial landscape.

2.12 Banks and Crypto-finance

Banks have the power to create the nation's money and credit, which they are entrusted to channel into productive economic uses, but the "money monopoly" and "money trusts" are one of the most dangerous forms of concentrated private wealth and a threat to economic freedom. Banks operate as the means by which monetary policy has an effect on the financial market, and so, bank regulation reduces the risks of intense competition,¹⁶⁶ but the concentration in banking accelerates concentration generally. It seems that formal bank mergers denote a dominant market position.¹⁶⁷ Hence, there is a need to apply all available anti-monopoly devices as part of a dynamic regulatory strategy. The government-granted monopoly on money creation rationalizes the bank regulatory regime's focus on conserving the public/private balance of power. Has this function been applied to cryptocurrencies circulation? It has to be taken into account that bank regulation and supervision prevent the over-issuance of money in relation to the needs and productive capacity of the nation's economy.¹⁶⁸ Has the emergence of cryptocurrencies and crypto assets averted/prevented the over-issuance of money? To that extent, S. T. Omarova and G. S. Steele¹⁶⁹ argue that "U.S. bank regulation operates as a comprehensive anti-monopoly regime, designed to prevent excessive concentration of private

power over the supply and allocation of money and credit in a democratic economy.”

It is characteristic that the continuing technology-driven shifts in the structure and operation of the financial system more and more call for a necessary modification in the paradigm of financial regulation.¹⁷⁰ Moreover, AI technological alterations in finance and the broader economy make the reconceptualization of banking law as an anti-monopoly structure a crucial part of any economic policy agenda.

It has to be taken into consideration that the digitization of financial services is not a value-neutral technological event; it is a political venture redefining financial and economic relationships in the AI era. Moreover, new participants modify the existing banking prototype and, in doing so, intensifying political-economy fears about anti-monopoly policy, which means that it is not merely containing the transaction-level technological revolution but administering changes in the distribution and exercise of structural authority in the financial markets.

So far, the advancement of fintech and crypto-finance has caused “shadow banking,” whereby money creation and credit allocation are reproduced outside of the regulated banking system. Hence, shadow banking in fintech and the crypto market is running around the existing banking rules and regulations, while evading the constraints of banking law.¹⁷¹ In other words, authorizing the growth of private forms of money and money substitutes intimidates the sovereign public’s capability to control the supply and flow of money and credit in the economy. Will states lose the power of money monopoly on behalf of a kind of financial NGOs operating in a “cloudy” environment and driving the globe on the road of merely accomplishing elite interests?

Most stablecoins are backed by US dollar reserves¹⁷² or other safe assets, and so, it seems that private digital currencies necessitate access to the nation’s full faith and credit. Hence, Stablecoins are collateralized by dollar bank deposits and government bonds from which they gain their value and capacity to perform as a private substitute for public money. Presently, dominant stablecoins are the central connecting means of crypto markets to the rest of the financial system. Thus, many crypto and fintech corporations link with regulated banks,¹⁷³ and so banks implement

depository and back-office services on behalf of tech platforms.¹⁷⁴ Hence, the amalgamation of digital money into the conventional system will allow control of the quantity of money in an economy. It seems that insecurely organized “money trusts” controlling the flows of money, credit, and commercial goods and services via their platforms will generate problems shortly, not only in the flow of funds but also in the stability of the financial market.

Also, it seems that banks are developing their own digital-asset operations, stimulating unparalleled growth of digital-asset markets. Therefore, tokenization offers the opportunity to relitigate the appropriate boundaries and applicability of existing rules to financial products tendered in new packages and with new labels, which means that the production of new formulas of structural interconnectedness between crypto corporations and regulated financial institutions raises substantial fears not only about financial stability but also about excessive concentrations of authority in digital platform-based finance.¹⁷⁵ Also, the conglomeration of tech platforms, data, and finance facilitate possibly systematic anticompetitive practices such as tying agreements among digital platforms.¹⁷⁶ It has to be taken into account that platforms take on anticompetitive discriminatory practices when they do not permit competitors to use their facilities under the ordinary terms offered to others.¹⁷⁷ On the other hand, tech platforms are common carriers under the common law, and accordingly, in common law, reasonable deplatforming may be accepted.¹⁷⁸ It is characteristic that antitrust analysis and enforcement have been diminished to the technical application of “microeconomic models.”¹⁷⁹

It is characteristic of the power of the Apple Pay digital wallet to restrict its tap-to-pay functions for disfavored goods, causing prospective implications for the applicability of open banking regulations. Additionally, tech finance platforms can engage in unfair pricing and manipulation of consumer behavior, utilizing competitor information by using their data and market power to entrap consumers.¹⁸⁰ It has to be taken into account that digital custodians offer a 500 billion USD digital-asset market under their terms and conditions for crypto custody and wallet services.¹⁸¹ To that extent, the MiCA rules on crypto custody link protection first to the custody of crypto assets, which then embraces the upholding of clients’ rights and

the avoidance of their reuse.¹⁸² Also, corporations are striving in the safeguarding and transacting of digital assets, whether it be from operational complexity, security, or reputational risks. Moreover, Zodia¹⁸³ Custody's Software-as-a-Service delivers bank-grade digital-asset technology services to customers by providing the flexibility and functionality required to succeed in a digital-asset economy.

It is characteristic that only a few custodians specify guidance to their customers on the applicable law and jurisdiction, and so, crypto has been distinguished by an absence of regulation involving contractual freedom, which means that financial regulation will be central to boosting investors' confidence in crypto and causing merit-based competition among providers. In line with this, D. Zetsche and A. Nikolakopoulou¹⁸⁴ argue that

“Many crypto custodians use misleading terminology, thus blurring the true extent of their accountability and liability with respect to their services. Moreover, the lack of transparency regarding the custody policies, and in particular the key storage mechanism, is a widespread practice in collective custody that carries serious risks for investors”.

In the end, many digital-asset custodians have been escalating their responsibilities from the safekeeping of cryptocurrencies to assisting customers in taking part in new business openings and asset classes, such as DeFi, NFTs, and metaverses.

2.13 Corruption and Tax Evasion

Corruption is acknowledged as a societal happening totally embedded in modern social interactions, escalating during transitional periods and encompassing practices like the sale of parliamentary seats, turning corruption into a tool for corrupt individuals within the system, which means that corruption illustrates a confrontation to institutional transformation. Hence, there is a need to promote transparency and accountability via a more effective procurement process. It has to be taken into consideration that corruption damages institutions, influencing the

credibility of states. Public and private sectors are part of corruption, and so, the public sector corruption denotes the misuse of public office for private benefits.

It is worth noting that corrupt dealings arise via a cost-benefit trade-off encompassing the tension between public agents' private interests and the general interest. Moreover, corporations' decisions to participate in corruption correspond to their objective of profit maximization, which means that corrupt dealings arise out of the dominance of profitability objectives over the societal value of ethics. Corruption in some societies denotes ethics and trust prevailing in their existence. To that extent, Chaudhry Anwar and Shabbir, Ghulam¹⁸⁵ argue that

“the socio-political and religious norms are so weak that they cannot affect the corruption level in these countries...

Therefore, perceived level of corruption is not affected by the religion...government should focus the economic factors to curb the level of corruption”.

For instance, corruption in the pharmaceutical system influences a state's entire population, and so, governments must establish sound institutional structures for implementing anti-corruption measures, processes, and policies in strengthening public welfare. Hence, corruption is reduced if institutions are transparent, public scrutiny is high, and the law and administrative processes are enforced properly by public officials. To that extent, Jillian Cohen¹⁸⁶ et al. write that

“The consequences of corruption within the pharmaceutical system are unfortunately fairly easy to identify. If quality control regulations are not sufficient or are not implemented or enforced, health and economic consequences result. At an extreme, unsafe counterfeit drugs can lead to severe health consequences, including death”.

Corruption is a substantial threat to national security, and the only way to successfully respond is to make certain the well-organized functioning of government bodies, strengthen national institutions, and considerably diminish the frequency of corruption.¹⁸⁷ Moreover, corruption and inefficiency in public services are related to weak governance and so illustrating the ethos and attitudes of the political and bureaucratic executive. Moreover, government policies, investment and financing decisions, customer-interfacing activities, commercial operations of the utilities, procurement, and human resource management are key fields prone to corruption. Moreover, corruption in procurement and contracting generates problems in many sectors of an economy, requiring an anti-corruption strategy concentrating on transparency and competition in procurement and contracting by externalizing accountability of public officials via public participation and independent regulatory institutions.¹⁸⁸ Also, corruption reroutes funds to projects with lower economic rates of return, and so, corrupted construction is dropping project sustainability and escalating the need for maintenance and rehabilitation.

It is worth noting here that combating corruption is about advancing governance. In fact, the implementation of AI systems dealing with electronic transactions means that AI can easily spot unusual transactions, which means that AI can contribute in dropping corruption. Government modernization via datafication and digitization offers the means for battling corruption because datafication and digitalization are central mechanisms via which regulatory managerialism functions.¹⁸⁹ Corporations participate in lobbying efforts to influence laws and trade with foreign countries. In line with this, Transparency International has criticized the absence of safeguards against lobbying, requiring international initiatives to stimulate public administration by stressing government transparency which means that there is a demand for coordinated efforts across all government entities throughout the countries and the presence of wide-ranging anti-corruption strategies.¹⁹⁰ Thus, merely establishing an anti-corruption agency is inadequate, and so, full commitment and cooperation from all government bodies is needed.¹⁹¹

It has to be taken into account that political institutions adjust the relationship between interest groups and policymakers and so generating

different patterns of mobilization and lobbying strategies. Corporations are adapting their lobbying strategies across different countries based on institutional differences.¹⁹² It seems that via lobbying access to decision-makers and influence on policies are traded against information and support. Also, lobbying is targeting legislators.¹⁹³ In line with this, J. Lee and J. Stuckatz¹⁹⁴ notify that

“interest groups that lobby across countries likely benefit from tailoring their lobbying strategies according to local institutions, and by strategically coordinating the activities of the parents and foreign subsidiaries. In fact, we find evidence that many multinational corporations that engaged in USMCA lobbying actually lobbied from both the US and Canada by utilizing their subsidiaries”.

Weak state administration that comes forward when an individual or organization has monopoly power over a good or service, discretion over making decisions, and limited or no accountability generates corruption, and so, individual economic success is not resting on market forces. It has to be taken into account that corruption accelerates economic growth by disabling unproductive allocation of resources stemming from weak institutions.¹⁹⁵ To that extent, Chaudhry Anwar and Shabbir, Ghulam¹⁹⁶ argue that

“increase in economic freedom, globalization and average level of income have reduced the level of corruption in these countries. But the level of corruption in developing countries is increased with the increase in level of education. The income distribution has not significantly explained the variations in the level of corruption for the countries in the sample”.

The quality of institutions advances the effectiveness of government expenditure and, in doing so, fostering economic growth. Naftaly Mose¹⁹⁷

argues that “the effect of government expenditure on growth is largely indirect through democracy and corruption. The results also show that the interaction of both democracy and corruption on government expenditure dampens economic growth.”

In general, corruption produces substantial distortions in the use of public resources to the detriment of society and so damaging democracies and the rule of law.¹⁹⁸ In other words, institutional corruption and conflicts of interest in tax law policymaking indicate the way that corruption affects policies against tax abuse, along with upsetting the rule of law and causing a distortion of justice. Hence, tax evasion and corruption are interrelated incidents causing harm on a global scale.¹⁹⁹ To that extent, corruption of tax officials is a substantial determinant of tax evasion. Additionally, tax havens, offshore centers, and free ports perpetrate tax evasion and tax avoidance.²⁰⁰ Regarding tax evasion offenses, L. Winter and D. Vozza²⁰¹ argue that

“a promising solution may lie in the introduction of a Convention Against Tax Evasion (CATE)...The convention would also offer the opportunity to send a clear message to corporate players and other sophisticated taxpayers who currently exploit the fragmentation of the international landscape about a new-found willingness to prevent and counter such illicit behavior”.

Globalization has transformed and intensified corrupt practices and tax evasion.²⁰² It has to be taken into account that tax evasion harms state revenue and necessary investments in society. There is no international legal mechanism responding to serious tax evasion and tax avoidance behaviors from a criminal law perspective.²⁰³

It seems that residual corruption, which is not linked with other governance indicators, advances economic growth in countries where the institutional quality is inadequate, and so, corruption clearly influences GDP per capita growth.²⁰⁴ Also, economic growth denotes a country’s macroeconomic stability. In other words, the powerlessness of weak

governments to inspect their institutions instigates a high level of corruption that harms growth irrespective of its impact on investment. Moreover, the substandard characteristics of institutions make the influence of corruption on growth more damaging.²⁰⁵ In line, Joël Cariolle²⁰⁶ denotes that

“Multi-level estimations stress that the negative effect of income per capita on bribery is found to be mostly driven by improvement in human capital, more particularly by the decline in fertility rates. They also allow the reconciling of some contrasting findings of the literature on other corruption determinants, but point that the contribution of corruption determinants is context-dependent”.

It has to be stressed here that the effect of corruption on growth is experienced some time prior to any variation is monitored in the corruption score.²⁰⁷ Moreover, the link between transparency and growth is direct with transparency leading to economic stability, greater investment, and growth.²⁰⁸

It is characteristic that corruption is an element triggering phenomena of “Arab uprisings”²⁰⁹ which means that corruption enhances economic growth by overpowering bureaucratic obstacles. In other words, corruption is utilized as an alternative to competitive bidding in purchasing government licenses, which also spreads bureaucratic slowness. Fatih Kırşanlı²¹⁰ mentions that

“The MENA region suffers from informal corruption practices such as *wasta* (networks and reciprocity) and *hamula* (clientelism). In the *wasta* system, citizens who have connections can quickly process their documents through the government bureaucracy and help their close ones get hired for a job”.

Given that political institutions greatly influence investment and growth, governments should exploit their resources to lessen corruption in political

and bureaucratic domains. It is worth noting that corruption acts as a strong impediment to economic activity, twisting public spending decisions and undermining the quality of infrastructure. To that extent, Fatih Kırşanlı²¹¹ mentions that “Eliminating corruption and boosting economic growth simultaneously will provide opportunities to lower corruption further, improve institutional quality, and sustain the economic development of each MENA country.” In general, it seems that corruption destabilizes the economic, physical, and social setting in which people make their production and consumption decisions, thus undermining factors of development such as the tangible infrastructure of a modern economy and the intangible capital.

Are corporations forced to follow corruption tactics in order to survive in a corrupt environment or to enter into a sovereign state? It seems that, on many occasions, corruption is good for an economy, permitting corporations and individuals to circumvent or expedite regulatory requests that are too costly or slow. Also, a capitalist who is balancing the costs and benefits of starting a business in a corrupt state calculates possible “bribes” they have to pay to those issuing licenses, to tax inspectors, etc., due to the fact that corrupt officials will affect profits and any generated intellectual property of the investment. Hence, capitalists and innovators in a corrupt country may not have much faith in the rule of law, which means that starting a business or investing in innovation in a corrupt country is riskier and carries extra costs.²¹² First of all, it seems that established corporations gain from corruption at the expense of the economy, particularly in order to survive in a corrupt setting when bribery and other kinds of corruption twist competition. Nonetheless, established corporations face extra costs from functioning in a corrupt environment, and thus, extra risks arise for them due to inadequate local knowledge and laws imposing noteworthy fines. Many corporations are not aware of the extent and depth of corruption until illegality or misconduct is revealed, and so, corruption interferes with the assessment of costs and benefits for new business establishments and investment decisions. In line with this, Robert Gillanders²¹³ refers that “corruption can grease the wheels of an economy that is subject to poor governance and overregulation...corruption is harmful to economic

development. While bribery may help some firms to overcome costly and inefficient regulation, corruption is a cause of red tape.”

It is worth noting that corruption may lead to projects that are of no benefit to the public, offering profitable contracts for connected individuals, which means that corruption may alter the composition of public expenditure, dropping trust. Also, the risk of corruption prevents innovation, hindering competition. In other words, corruption diminishes the extent and competence of public and private investment, prohibiting beneficial trades.

As mentioned earlier, corruption is a means of employing public powers to accomplish private objectives. Moreover, corruption takes place when there is a monopoly of power, restricted responsibility, and power to make judgments. To that extent, legal reforms and social changes reduce corruption. Will AI spot and detect properly corruption tactics? To that extent, AAI systems can be employed in monitoring matters of corruption and its impact on investment and growth, thereby permitting an automatic identification and fixing of any mismatch on behalf of competitive implementation. In line with this, Joël Cariolle²¹⁴ denotes that

“multi-level estimates support that economic development, measured by income per capita, significantly and negatively contributes to bribery prevalence...Last, estimations emphasize the direct and indirect effects of democratic institutions on bribe prevalence, and also stress contrasting evidence: while increased political rights and media independence have, as expected, a significant deterrent effect on corrupt transactions, greater civil liberties are found to foster bribery”.

Hence, the introduction of AAI technology running objectively will spot easily matters of corruption and will automatically repeal any transaction grounded in corruption and not in the rule of law.

On the other hand, in certain contexts, corruption augments efficiency by upsetting the scales of cost, advancing economically beneficial activities. In

other words, corruption is beneficial in “greasing the wheels” of the economy in that way stimulating competence and effectiveness in the economy, particularly for developing countries with weak institutions.

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